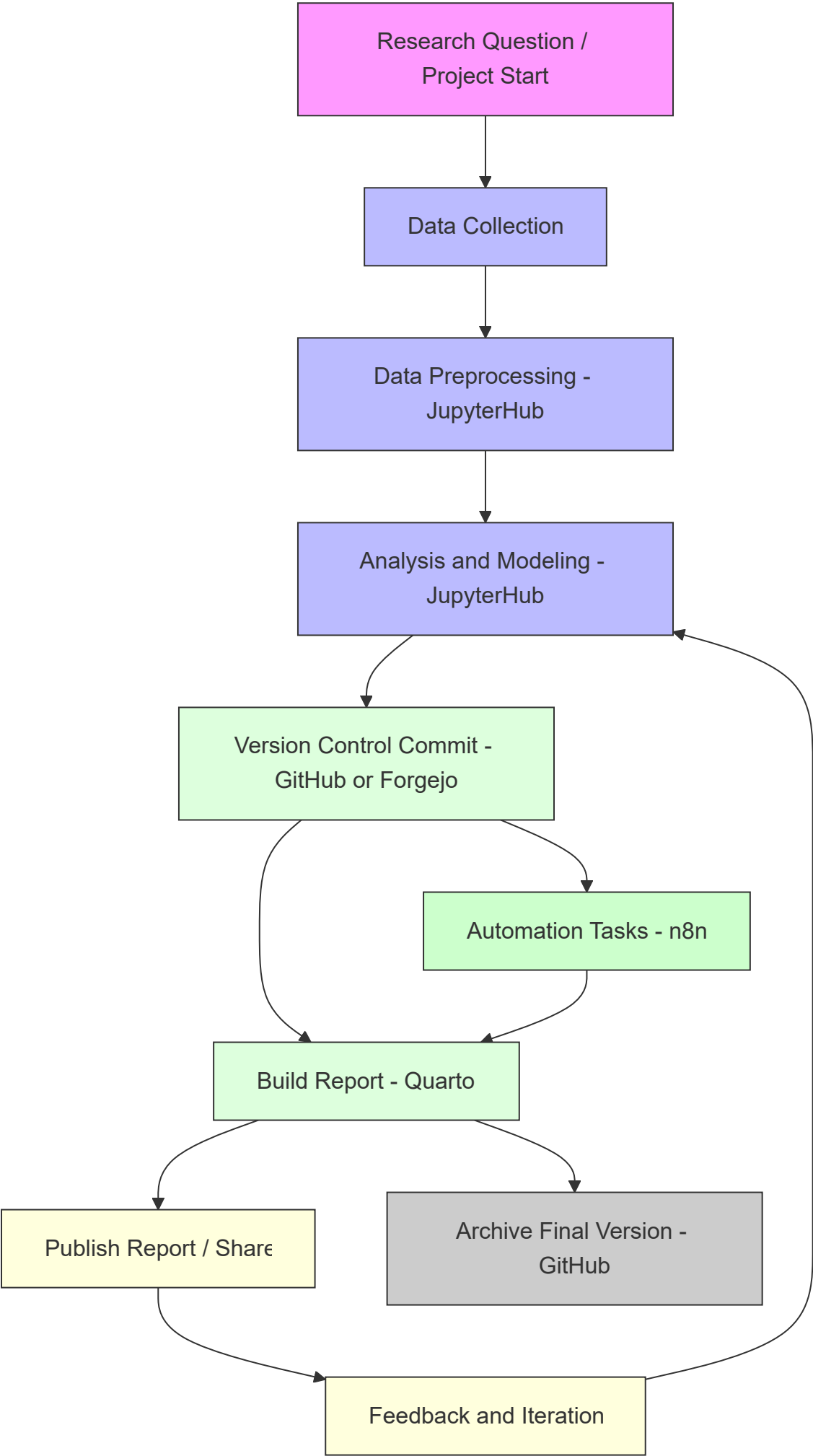


Reproducible Research Infrastructure for the Value Chain Hackers Program

Role Context: *Value Chain Hackers (VCH)* is an applied research program dedicated to tackling complex supply chain challenges through innovation and collaboration. The program’s mission is to “*revolutionize the supply chain industry through relentless innovation, spirited collaboration, and hands-on practical application*”, engaging both academic researchers and industry partners in joint projects. In this assignment, you (as a graduation-level researcher in VCH) will take on the role of a lead architect tasked with ensuring that VCH’s research process is fully **reproducible** – meaning that analyses, data handling, and results can be reliably repeated and audited by others. Reproducibility is crucial in VCH’s academic–industry collaborations because it builds trust, enables knowledge transfer between project teams, and allows future researchers to build upon past work with confidence. This brief outlines your assignment objectives, scope, deliverables, and evaluation criteria as you design and prototype a **reproducible research infrastructure** using the *Enterprise Data Reference Model (EDRM)* (suggestion) as a guiding framework.

Objective

The primary objective of this assignment is to **investigate, implement, and document a reproducible research process** for the VCH program, using principles from the Enterprise Data Reference Model. In practical terms, you will design an end-to-end workflow that integrates with VCH’s existing infrastructure and tools, ensuring that research activities are *scalable*, *reusable*, and *transparent* across future projects. This means establishing standard ways to collect, organize, analyze, and share data and findings so that any qualified person can follow the process and achieve the same results. By leveraging the EDRM framework – a set of standards for consistent, mission-aligned data management – you will bring structure and quality assurance to VCH’s research workflows. Ultimately, you are expected to produce a working prototype of this infrastructure and comprehensive documentation that together demonstrate how VCH research projects can be conducted in a reproducible manner from start to finish.



Scope of Work

Your work will involve several key activities to build the reproducible research infrastructure. **Scope of work** includes, but is not limited to, the following steps:

1. **Map Current Research Workflows:** Examine and diagram the way VCH research is currently conducted. This includes identifying how data is collected, processed, and analyzed in ongoing projects, what tools are used, and how results are documented. The goal is to understand the *as-is* workflow and pinpoint any inconsistencies or informal practices.
2. **Identify Reproducibility Pain Points:** Analyze the current workflows to find where reproducibility might fail. For example, note if there are manual data transformations that aren't documented, results that vary between runs, or collaboration points where information gets lost. These failure points (e.g. lack of version control, undocumented assumptions, inconsistent data formats) will inform what needs improvement in the new design.
3. **Apply EDRM Principles to Workflow Design:** Using the Enterprise Data Reference Model as a guide, restructure the research workflow for consistency and clarity. This may involve defining standard data schemas or metadata, establishing how datasets and results are stored and described, and enforcing quality standards at each step. The EDRM's emphasis on **common data definitions and governance** should help align the research process with best practices in data management. At this stage, design the architecture of the new workflow (e.g. what components it has, how data and results flow through it, and how each step is documented).
4. **Prototype with VCH-Infra Tools:** Implement a working prototype of the new reproducible workflow using VCH's existing infrastructure:
 - **JupyterHub** – for collaborative Jupyter notebooks, allowing multiple team members to perform analysis in a shared, controlled environment.
 - **Quarto** – for creating dynamic, publication-quality documents and reports directly from notebooks and Markdown, ensuring that documentation and results are automatically updated when the underlying code or data changes.
 - **GitHub (or Forgejo)** – for version control of all project assets (data, code, notebooks, documentation). This repository will track changes and facilitate collaboration, and can serve as a template for future projects.
 - **n8n** – for workflow automation and orchestration (e.g. scheduling notebook runs, automating data pipeline steps, or integrating different services). Using n8n, you might automate tasks such as data refreshes, analysis reruns, or notifications, embedding reproducibility into the process by minimizing manual intervention.

These tools will be made available to you in the 2nd week of your starting assignment.

In this prototyping phase, you will set up and configure these tools to work together. For instance, a possible prototype could involve an automated pipeline where n8n triggers a JupyterHub notebook to run analysis on new data, pushes results to GitHub, and Quarto renders an updated report.

5. **Documentation in Quarto:** Develop thorough documentation of the infrastructure and methods. Using Quarto, create a well-structured document or site that explains the workflow architecture, how to use the prototype, and the underlying methodology. This Quarto documentation should serve both as a user guide and as an academic report: include diagrams of the workflow, descriptions of how data flows through the system, and rationale for design decisions (especially how EDRM principles are implemented). The documentation should itself be reproducible – for example, including embedded results or charts that are generated from the latest data via the pipeline.

Throughout these activities, ensure to test and refine the workflow. You may use a small case study (for example, a sample supply chain dataset or a prior VCH project) to validate that someone else could follow your documented process and get the same results. The scope encompasses both the technical implementation and the methodological framework, so devote attention to both how to do things (tools and code) and why they are done that way (principles and standards).

Expected Deliverables

By the end of this assignment, you are expected to produce several **deliverables** that demonstrate the successful design and implementation of the reproducible research infrastructure:

- **Reproducible Workflow Prototype:** A working prototype of the research workflow, integrating JupyterHub, Quarto, Git/GitHub (or Forgejo), n8n, and any other necessary tools. This should be configured and demonstrated with example data – for instance, a notebook that runs end-to-end, yielding a result that can be independently verified. The prototype should show how a future VCH research project could be conducted using your infrastructure.
- **Quarto Documentation of Architecture & Methodology:** A comprehensive document (or website) created with Quarto that describes the entire infrastructure. This should include the system architecture diagram, explanations of how each component (JupyterHub, Quarto, etc.) contributes to reproducibility, and instructions for users on executing and extending the workflow. It should also detail how EDRM concepts were applied (e.g. standard data definitions, metadata practices) to enhance reusability and clarity.
- You will receive extensive support from Christiaan Verhoef in the implementation of this, you are not expected to be technically adept for this internship.
- **Written Reflective Report:** A report (which might be part of the Quarto document or separate) in which you reflect on the process and context. This report should discuss any **institutional or organizational barriers** to reproducible research you encountered or foresee (for example, cultural habits, lack of policies, data silos) and identify opportunities for improving reproducibility at the institutional level. Essentially, it should provide a critical analysis of what challenges arose when embedding reproducibility in VCH and how these challenges can be overcome, as well as the broader benefits to the program or university.
- Ideally this report should be done in a manner such that you show that you have applied this methodology yourself.
- **Version-Controlled Repository:** A public or private repository (on GitHub) containing all code, data (or data references), configuration files, and documentation for the project. The repository should be organized logically (using README files, folders, and clear naming) and should

demonstrate good version control practices (meaningful commit history, use of branches or tags if appropriate). This deliverable will illustrate the *reusable template* aspect of your work – future students or researchers should be able to fork or copy this repository to jump-start their own reproducible research projects in VCH.

Each deliverable will be evaluated for completeness and quality. For example, the prototype should be not only functioning but also clearly reproducible by someone else following your instructions. The documentation should be clear and sufficient for a newcomer to understand and use the system. The reflective report should be insightful and well-supported. Together, these deliverables will form a package that equips VCH with a robust reproducible research infrastructure for the long term.

Why This Matters

Building a reproducible research infrastructure is strategically important for the VCH program and its stakeholders, beyond just completing a one-off project. Here are a few key reasons *why reproducibility is critical in this context*:

- **Continuity:** VCH operates in an academic setting with high turnover (students graduate, projects conclude). A reproducible process ensures *continuity* of knowledge, new team members or future research cohorts can pick up where previous ones left off. By standardizing workflows and documenting them thoroughly, you enable long-term projects that outlast individual participants. This prevents loss of valuable work and facilitates cumulative progress over multiple semesters.
- **Institutional Impact:** By creating a template for reproducible research, you contribute to the institutional knowledge and capacity of VCH and the university at large. Well-documented, repeatable research processes can be adopted as best practices program-wide, improving the quality of all VCH projects. This alignment with rigorous standards (such as EDRM and open-science principles) enhances the *credibility and impact* of VCH's output. In the eyes of both academia and industry, the institution gains a reputation for delivering reliable, verifiable results, which can open doors for further collaborations and funding.
- **Accountability and Transparency:** Reproducible research fosters transparency, which is vital when working with industry partners or any external stakeholders. If findings and analyses can be audited step-by-step – with clear provenance of data and methods – stakeholders can trust the outcomes. This accountability means that decisions or innovations derived from VCH research are backed by evidence that others have validated. It also helps internally: teams can more easily review each other's work, ensuring higher quality and catching errors early. Ultimately, a culture of transparency and accountability strengthens the partnership between the academic team and industry collaborators by ensuring everyone can see *how* results were obtained.
- **Scalability and Reusability:** Designing the process to be reproducible inherently makes it more scalable and reusable. When each project follows a core template, successful methods can be repeated and scaled up to larger problems or adapted to new contexts with minimal friction. This consistency reduces the time needed to start new projects (since the infrastructure is in place) and allows parallel teams to collaborate or share components of their workflows. It creates an ecosystem where improvements in the process by one team can benefit all teams.
- **Stopping bullshit before it spreads:** If you cannot trace back how results were made, bad data and nonsense spread fast. As McGonigal points out, once the story gets told the wrong way, people believe it. Then you spend more time fighting the lie than solving the problem. If we make our research traceable, others can check it and fix mistakes early. This protects the value of what we do and keeps VCH from being part of the problem.

Timeline

The assignment should be completed over a series of phases. Below is an indicative **timeline** outlining stages of work, which you can adjust as needed. Each stage corresponds to a focus area of the project:

1. **Research & Planning (Weeks 1–2):** Conduct preliminary research into current VCH workflows and the EDRM framework. Gather requirements by talking to team members and reviewing existing project documentation. Define success criteria for the new infrastructure. *Output:* workflow diagrams of the current state, a list of reproducibility issues, and a high-level plan for the new system.
2. **Design & Prototype Development (Weeks 3–6):** Apply EDRM principles to design the new workflow and begin prototyping. Set up the tools (JupyterHub environment, Git repository, n8n instance, etc.) and start building the pipeline. Implement small proof-of-concept tests for each component (e.g. a sample notebook running through JupyterHub and being rendered by Quarto). *Output:* an initial version of the reproducible workflow, possibly tested on a simple example, and an outline of the documentation.
3. **Validation & Iteration (Weeks 7–8):** Test the prototype with realistic scenarios or data. Run through the entire process as if executing a real research project. Validate that another person can follow the documented steps to reproduce results. Identify any bugs, pain points, or deviations from reproducibility standards and refine the system. This stage may involve getting feedback from peers or mentors on usability. *Output:* a refined workflow that has been proven to work end-to-end, and notes on any adjustments made for reliability.
4. **Documentation & Final Reporting (Weeks 9–10):** Finalize the Quarto documentation and the written report. Ensure the documentation includes all necessary instructions and reflects the latest state of the prototype. Complete the reflective analysis of barriers and opportunities, incorporating any insights gained during development and testing. Prepare the repository with clear instructions (e.g. an updated README) for submission. *Output:* the finished Quarto documentation (published as HTML/PDF if required), the completed report, and a polished repository ready for review.
5. **Presentation & Submission (Week 11):** Although not a separate deliverable in writing, plan to present your reproducible infrastructure to stakeholders (faculty advisors, possibly industry mentors) as part of the evaluation. This typically involves a demonstration of the working prototype and a summary of your findings and recommendations. *Output:* presentation slides or demo notebooks, and the formal submission of all deliverables for grading.

Assessment Criteria

Your work will be evaluated on several **criteria** to ensure both the technical and analytical goals of the assignment are met. The following are key areas that will be assessed:

- **Technical Quality of Implementation:** The robustness and effectiveness of your prototype infrastructure. This includes how well the tools (JupyterHub, Quarto, GitHub/Forgejo, n8n) are configured and integrated, and whether the workflow actually achieves reproducibility (e.g. producing the same results consistently on re-run or on a different machine). Use of the EDM principles will be checked here – for example, did you implement standards for data handling and did those improve the workflow? A high-quality implementation will be reliable, efficient, and aligned with the best practices discussed in the course.
- **Reproducibility & Sustainability:** The degree to which your research process can be *repeated and reused*. A reproducible workflow should yield identical results given the same inputs, and your documentation and code should enable others to replicate the analysis without ambiguity. Sustainability refers to how well the solution can continue to be used or adapted in the future. This involves the clarity of organization (for someone inheriting the project) and use of maintainable technologies or configurations. In essence, we will assess if your infrastructure is not just a one-time demo, but something that could be adopted for real ongoing projects in VCH.
- **Documentation Clarity and Completeness:** The quality of your Quarto documentation and any supplementary materials. Clarity is paramount – the documentation should be understandable to other students and researchers in VCH. It must clearly explain how to execute the workflow and why it is designed that way. Completeness means all aspects of the system are documented: setup instructions, usage examples, explanations of data structures, etc. The documentation should also be well-structured (using appropriate sections, headings, diagrams, and references) and written in a professional, concise manner. Remember, this documentation doubles as a user guide and as proof of your methodology, so it will be judged on both technical accuracy and writing quality.
- **Critical Analysis & Reflection:** The depth and insight of your written report on institutional barriers and opportunities for reproducible research. We will evaluate how well you identified challenges (e.g. cultural, technical, or organizational issues that make reproducibility hard) and the realism and creativity of your proposed solutions or recommendations. This criterion looks at your ability to connect the practical work with broader concepts – for instance, discussing how the adoption of your infrastructure might change VCH’s research culture, or what support structures are needed to maintain it. Strong reflection will demonstrate learning from the project and an understanding of the *bigger picture* of reproducible science in an enterprise context.
- **Adherence to Requirements:** Finally, your submission will be checked against the assignment requirements and scope. This means all expected deliverables should be present and functional (prototype, documentation, report, repository), and the project should remain within the scope defined (focusing on VCH’s needs and using the specified tools and framework). Meeting deadlines and producing a well-organized submission are also part of professional execution. While this is more of a binary criterion, failing to include a key component (for example, no reflection report or missing documentation sections) would affect your grade.

Each of these criteria will have an assigned weight in the grading rubric, which will be shared separately. To excel, aim for a balanced performance: a brilliant technical solution without clear documentation, or a great report without a working prototype, will not fulfill the assignment’s purpose. We encourage you to use the feedback from mentors and peers throughout the project to improve in all these areas before final submission.

Stakeholders and Collaboration

This project will not be done in isolation – part of your role involves engaging with various **stakeholders** in and around the VCH program to ensure the solution is well-rounded and adoptable:

- **Faculty Advisors:** You will work under the guidance of faculty mentors (such as the VCH program leaders or your academic supervisor). They are available for consultation at planning stages and for feedback on your design. Regular updates to your advisor will help ensure your approach aligns with academic standards and VCH’s strategic goals. Faculty will ultimately evaluate your work, so treating them as both stakeholders and evaluators is wise incorporate their feedback on what a successful infrastructure should achieve.

Here are some of the mentors that will be available to you:

Christiaan Verhoef - Windesheim Zwolle
 Fred van toor - Program coordinator Finance - Windesheim Almere
[Josef Z. Koehl](#)[View Josef Z. Koehl's profile](#) - Viridis
 Patrick Burgess - Program Coordinator of Master's (MSc) Agribusiness Developmen
 Alfred Pot - Manager Techniek - Aeres

- **Peer Researchers and Student Teams:** The reproducible infrastructure you develop is intended to be used by other student teams in VCH. Consider involving one or two fellow students (or a small team) to pilot-test your workflow. Their perspective will help gauge usability: for example, can they follow your instructions without your help? Engaging peers through demos or short user testing sessions can uncover pain points you might not see. Additionally, collaboration with your cohort (through Scrum meetings or workshops) will ensure your project stays relevant to the needs of the teams who will use it next.
- **Industry Advisors/Partners:** There are 2 rounds of presenting during the entire project, in which the researchers that you advice will need to present to VCH's industry partners like Scania.